

ORF 309 – Homework 4, Q2

Q2(a): Probability exactly half the class has COVID during the first lecture

Setup. There are $n = 140$ students. Each student independently has COVID on a given day with probability $p = 0.005$, so $q = 1 - p = 0.995$.

Let X_i be the Bernoulli RV for student i :

$$X_i = \begin{cases} 1 & \text{if student } i \text{ has COVID} \\ 0 & \text{otherwise} \end{cases}, \quad P\{X_i = 1\} = p, \quad P\{X_i = 0\} = q$$

Let $S_n = X_1 + X_2 + \cdots + X_n$ be the total number of students with COVID. Since the X_i are independent Bernoulli(p) trials:

$$S_n \sim \text{Binom}(n, p) = \text{Binom}(140, 0.005)$$

“Exactly half the class” means $S_n = 70$. Applying the Binomial PMF:

$$P\{S_n = 70\} = \binom{140}{70} p^{70} q^{70} = \binom{140}{70} (0.005)^{70} (0.995)^{70}$$

This is essentially 0. The expected count is $E[S_n] = np = 140 \times 0.005 = 0.7$, so 70 is wildly far from the mean.

Q2(b): Distribution of first week with COVID during at least one lecture

Step 1: Probability at least one student has COVID on a single lecture day.

On any lecture day, $S_{140} \sim \text{Binom}(140, 0.005)$. Using the complement:

$$\begin{aligned} P\{S_{140} \geq 1\} &= 1 - P\{S_{140} = 0\} \\ &= 1 - \binom{140}{0} p^0 q^{140} \\ &= 1 - q^{140} = 1 - (0.995)^{140} \end{aligned}$$

Step 2: Probability at least one lecture in a week has a COVID case.

There are 3 lectures per week. Define events for each day:

$$D_1 = \{\text{at least one student has COVID on Monday}\}$$

$$D_2 = \{\text{at least one student has COVID on Wednesday}\}$$

$$D_3 = \{\text{at least one student has COVID on Friday}\}$$

Let $A = D_1 \cup D_2 \cup D_3 =$ at least one lecture day has a COVID case. It is easier to compute the complement $A^c = D_1^c \cap D_2^c \cap D_3^c$ (all three days are COVID-free). Since the days are independent:

$$P\{A^c\} = P\{D_1^c\} \cdot P\{D_2^c\} \cdot P\{D_3^c\} = [q^{140}]^3 = q^{420} = (0.995)^{420}$$

$$P\{A\} = 1 - P\{A^c\} = 1 - q^{420} = 1 - (0.995)^{420}$$

Step 3: Model the first week as T_1 in a Bernoulli process.

Think of each week as a Bernoulli trial with “success” probability

$$\tilde{p} = P\{A\} = 1 - (0.995)^{420}$$

The weeks are independent, so we have a Bernoulli process with parameter $\tilde{p}$. Let T_1 = the first week with a COVID case during at least one lecture. By the Geometric distribution (Slides 08):

$$T_1 \sim \text{Geom}(\tilde{p}), \quad \tilde{p} = 1 - (0.995)^{420}$$

with PMF $P\{T_1 = w\} = \tilde{p}(1 - \tilde{p})^{w-1}$ for $w = 1, 2, 3, \dots$

Numerically: $(0.995)^{420} \approx 0.1225$, so $\tilde{p} \approx 0.8775$ and $E[T_1] = 1/\tilde{p} \approx 1.14$.

Appendix: The Geometric Distribution

What it models

The Geometric distribution answers one question: **how many independent trials until the first success?**

You have a Bernoulli process — a sequence of independent trials $X_1, X_2, X_3, \dots$ where each trial succeeds with probability p and fails with probability $q = 1 - p$. Define:

$$T_1 = \min\{n \geq 1 : X_n = 1\} = \text{trial number of first success}$$

Deriving the PMF from scratch

For $T_1 = w$ (first success on trial w), you need a very specific sequence of outcomes:

$$\underbrace{X_1 = 0, X_2 = 0, \dots, X_{w-1} = 0}_{w-1 \text{ failures}}, \quad X_w = 1$$

Since all trials are independent, “and” becomes multiplication:

$$\begin{aligned} P\{T_1 = w\} &= P\{X_1 = 0\} \cdot P\{X_2 = 0\} \cdots P\{X_{w-1} = 0\} \cdot P\{X_w = 1\} \\ &= \underbrace{q \cdot q \cdots q}_{w-1 \text{ times}} \cdot p \\ &= q^{w-1} p \end{aligned}$$

This is the Geometric PMF: $\boxed{P\{T_1 = w\} = q^{w-1} p, \quad w = 1, 2, 3, \dots}$

Verification: probabilities sum to 1

$$\sum_{w=1}^{\infty} q^{w-1} p = p \sum_{w=1}^{\infty} q^{w-1} = p \cdot \frac{1}{1-q} = \frac{p}{p} = 1 \quad \checkmark$$

(Used the geometric series $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ for $|r| < 1$.)

Expected value: $E[T_1] = 1/p$

Intuition: If each trial succeeds with probability p , on average you need $1/p$ trials. If $p = 0.5$ (fair coin), you expect 2 flips to get heads. If $p = 0.01$, you expect 100 trials.

Derivation:

$$E[T_1] = \sum_{w=1}^{\infty} w \cdot q^{w-1} p = p \sum_{w=1}^{\infty} w q^{w-1}$$

Using the identity $\sum_{w=1}^{\infty} w r^{w-1} = \frac{1}{(1-r)^2}$ (derivative of geometric series):

$$E[T_1] = p \cdot \frac{1}{(1-q)^2} = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

Concrete example

Roll a fair die until you get a 6. Each roll has $p = 1/6$, $q = 5/6$.

- $P\{T_1 = 1\} = \frac{1}{6} \approx 0.167$ (get a 6 on first roll)
- $P\{T_1 = 2\} = \frac{5}{6} \cdot \frac{1}{6} \approx 0.139$ (miss, then 6)
- $P\{T_1 = 3\} = \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} \approx 0.116$ (miss twice, then 6)
- $E[T_1] = 6$ rolls on average

How it was used in Q2(b)

In Q2(b), we aren't flipping coins — we're asking “does this week have a COVID case during a lecture?” But the structure is identical:

- Each week is a “trial”
- “Success” = at least one COVID case during a lecture ($\tilde{p} = 1 - q^{420}$)
- “Failure” = no COVID cases all week ($1 - \tilde{p} = q^{420}$)
- Weeks are independent

So T_1 = first week with a COVID case follows $\text{Geom}(\tilde{p})$. The Geometric distribution applies whenever you have independent repeated trials and ask “when does something first happen?”